

CHAPTER 14

Digital-to-Analog Converters

LEARNING OBJECTIVES

Upon completion of this chapter on digital-to-analog converters, you will be able to:

- Write the general input–output equation for a digital-to-analog converter (DAC) and calculate the output for any given input.
- Understand basic DAC specifications.
- Distinguish between a DAC and an MDAC.
- Tell what features must be present to make a DAC compatible with a microprocessor.
- Explain how a microprocessor selects only one DAC out of all peripheral devices and sends data to it.
- Dynamically test a microprocessor-compatible DAC, the AD558.
- Explain the operation of a serial DAC and how to program a microprocessor to send data to it.

14-0 INTRODUCTION

Real-world processes produce analog signals that vary continuously. The rate may be very slow, such as a change in temperature variations, or very fast, such as in an audio system. Analog processes are best described by decimal numbers and letters of the alphabet. Microprocessors and computers, however, use binary patterns to represent numbers, letters, or symbols.

It is not easy to store, manipulate, compare, calculate, or retrieve data with accuracy using analog technology. However, computers can perform these tasks quickly and on an almost unlimited mass of data with precision, using digital techniques. Thus the need for converters to interface between the analog and digital worlds emerged. Analog-to-digital converters (ADCs) allow the analog world to communicate with computers. Computers communicate with people and physical processes via digital-to-analog converters (DACs). This chapter and the next study this interconnection between the analog and digital worlds. This chapter concentrates on digital-to-analog converters while Chapter 15 covers the analog-to-digital interface. The reason for studying digital-to-analog converters first is that some analog-to-digital converters use DAC circuitry as you will see in Chapter 15.

14-1 DAC CHARACTERISTICS

A digital-to-analog converter converts an input binary pattern of logic 1s and 0s to a discrete analog output, either voltage or current. The input–output relationship for a DAC is a set of points corresponding to the input digital code and its equivalent analog output. This relationship is the DAC's *transfer function*. Depending on how the DAC is configured, the transfer can be unipolar (only positive output values) or bipolar (positive and negative values). DACs can be purchased with an analog output voltage or with a current output. Current output DACs require the user to add an op amp to the DAC's output terminal. However, this allows the user to select the best device and set the circuit gain for the application. This section describes some of a DAC's specifications.

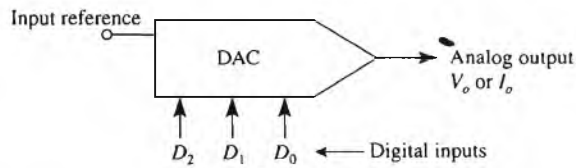
14-1.1 Resolution

The circuit symbol and the input–output characteristics of a 3-bit DAC are shown in Figure 14-1(a). Each digital input requires an electrical signal representing either a logic 1 or a logic 0. D_0 is the least significant bit, LSB, and for this example, D_2 is the most significant bit, MSB.

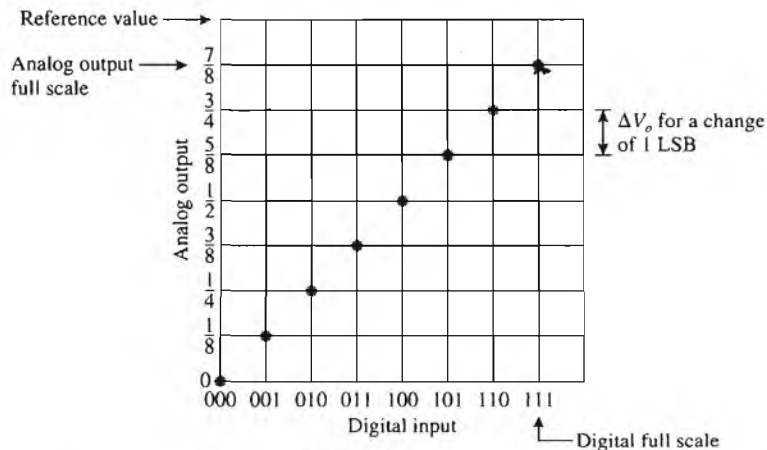
Resolution is the number of distinct analog outputs (voltage or current) that can be produced by a DAC. For an n -bit DAC,

$$\text{resolution} = 2^n \quad (14-1)$$

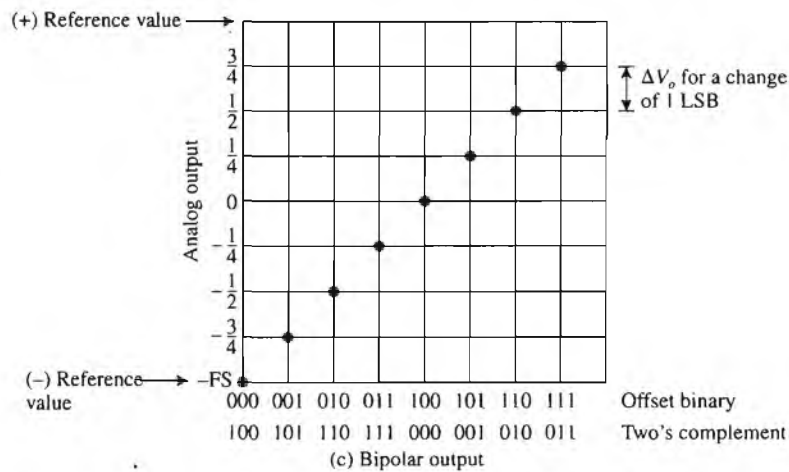
where n is the number of digital inputs. Therefore, for the 3-input DAC of Fig. 14-1(a), there are $2^3 = 8$ distinct outputs. Figure 14-1(b) is a plot of the discrete analog outputs for the corresponding digital input codes. This plot is referred to as the DAC's transfer



(a) DAC circuit symbol



(b) Unipolar output



(c) Bipolar output

FIGURE 14-1 (a) Circuit symbol; (b) and (c) plot of a 3-bit DAC's transfer function for unipolar and bipolar output.

function. Figure 14-1(b) is a plot for unipolar operation (all output values are positive). Figure 14-1(c) is a plot of a transfer function for a 3-bit DAC wired for bipolar operation. The digital input code for bipolar operation may be either offset binary or two's complement. Both codes are shown in Fig. 14-1(c).

The resolution value is used to determine the change in analog output voltage, ΔV_o , resulting from a change of 1 LSB at the digital input. See Figs. 14-1 (b) and (c). In order to determine this change, we need to know the DAC's full-scale output range and its resolution. In equation form,

$$\Delta V_o = \frac{\text{full-scale range}}{2^n} \quad (14-2)$$

Consider the 3-bit DAC of Fig. 14-1(a) with the input reference set at 1.2 V. Then from Eq. (14-2), the change of the output voltage for a change of 1 LSB is 150 mV/bit (1.2 V/8).

Note: A digital code equal to the full-scale range (FSR) does not exist. However, the industry standard positive full scale is the full-scale range less 1 LSB and is given by

$$V_{fs} = V_{ref} \left(1 - \frac{1}{2^n} \right) \quad (14-3)$$

where V_{fs} is the ideal full-scale output voltage when the digital inputs are all 1s. V_{ref} is the output reference value, which is usually equal to a multiple of the input reference. For our applications, V_{ref} equals the input reference voltage. This value may be set by a voltage reference IC as described in Section 2-6.

Example 14-1

An 8-bit DAC wired for unipolar operation has a full-scale voltage range from 0 to 5.12 V. What is (a) the DAC's resolution, (b) the output voltage change per bit, and (c) the ideal full-scale output voltage when all logic 1s are applied at the input?

Solution

(a) Applying Eq. (14-1),

$$\text{resolution} = 2^8 = 256$$

(b) The output change per bit is the output full-scale range (the reference value) divided by the resolution as given by Eq. (14-2).

$$\Delta V_o = \frac{5.12 \text{ V}}{256} = 20 \text{ mV/bit}$$

(c) Applying Eq. (14-3),

$$V_{fs} = 5.12 \text{ V} \times \left(1 - \frac{1}{2^8}\right) = 5.1 \text{ V}$$

Note: The full-scale output is always 1 LSB less than the maximum full-scale voltage range (the reference value).

Example 14-2

An 8-bit DAC has a resolution of 20 mV/bit. What is the analog output voltage for the following digital input codes (the MSB is the leftmost bit)?

(a) 00010110

(b) 10000000

Solution One method is to obtain the decimal equivalent value of the binary pattern and then multiply the decimal value by 20 mV/bit. The binary weighted code for unipolar operation is

$$D = b_7 2^7 + b_6 2^6 + b_5 2^5 + b_4 2^4 + b_3 2^3 + b_2 2^2 + b_1 2^1 + b_0 2^0$$

where D is the decimal equivalent value.

(a) Applying the binary weighted code yields

$$D = 0 + 0 + 0 + (1)2^4 + 0 + (1)2^2 + (1)2^1 + 0 = 22$$

Then

$$V_o = 22 \times 20 \text{ mV/bit} = 0.44 \text{ V}$$

(b) Applying the binary weighted code yields

$$D = (1)2^7 + 0 + 0 + 0 + 0 + 0 + 0 + 0 = 128$$

Then

$$V_o = 128 \times 20 \text{ mV/bit} = 2.56 \text{ V}$$

Example 14-3

If the DAC of Example 14-1 is wired for bipolar operation, what is the change in output voltage per bit?

Solution For bipolar operation, the full-scale output voltage range is twice the unipolar reference value. Hence,

$$\Delta V_o = \frac{2 \times V_{ref}}{2^8} = \frac{2 \times 5.12 \text{ V}}{256} = 40 \text{ mV/bit}$$

For bipolar operation, each bit change causes twice the output change as unipolar operation. Compare Fig. 14-1(c) with Fig. 14-1(b).

Typical DAC resolutions are 8, 10, 12, 14, 16, 18, 20 bits. However, DACs with higher resolutions are available. A DAC's digital input circuitry is designed for TTL, ECL, or CMOS voltage levels and this specification should be checked to match your application.

14-1.2 Offset Error

The transfer functions of Fig. 14-1 are ideal. They do not show errors that occur in practical devices. Figure 14-2 shows a plot of a transfer function due to an offset error of one LSB. Although this plot is for a positive offset error, the error could have been negative. For most devices, the offset error is usually measured by applying an input digital code of all logic 0s and measuring the analog output—the deviation from zero output is the offset error. *Note:* An offset error does not always occur just at zero. The reason that the offset error is measured at zero is because there usually are no other errors at zero. However, in some DAC this is not so, and the user should check with the manufacturer's data sheet to see if other techniques have been used. Figure 14-2(b) shows an offset error range, positive and negative, for a DAC wired for bipolar operation.

Example 14-4

A 12-bit unipolar DAC has a maximum offset error specified as 0.05% of FSR. The input reference voltage is set at 10.24 V. (a) What is the offset voltage? (b) What is the offset value in terms of least significant bit(s)?

Solution (a)

$$\text{Offset voltage} = 0.05\% \times 10.24 \text{ V} = 5 \text{ mV}$$

(b) In terms of least significant bit(s)

$$\Delta V_o = \frac{10.24 \text{ V}}{2^{12}} = 2.5 \text{ mV/bit}$$

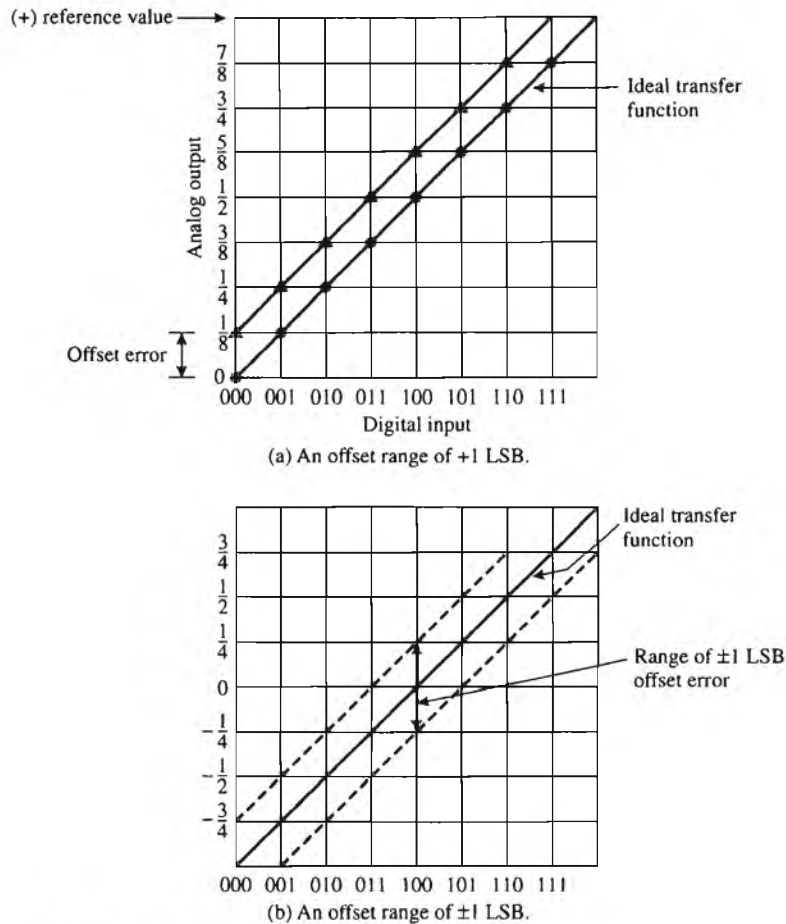


FIGURE 14-2 Offset errors.

$$\therefore \frac{\text{Offset voltage}}{\Delta V_o} = \frac{5 \text{ mV}}{2.5 \text{ mV/bit}} = 2 \text{ LSBs}$$

Although this example specified only a positive offset error, offset errors are usually specified as $\pm$ percentages of FSR.

14-1.3 Gain Error

Another error that can exist in the output of a DAC is gain error. Figure 14-3 shows the result of gain error. The slope of the transfer function differs from the ideal case. In the ideal case, the slope can be determined by drawing a straight line through the end points.

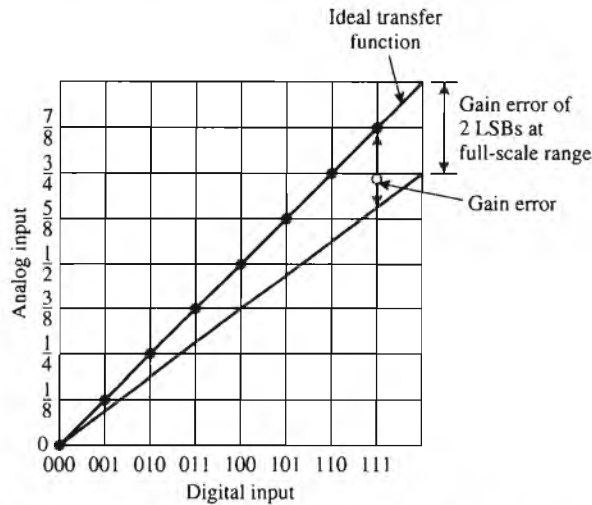


FIGURE 14-3 A gain error of 2 LSBs at full-scale digital input.

The slope is the gain of the transfer function. Figure 14-3 shows the effects of gain error and how it differs from the ideal transfer function. Gain error is given as a percentage of the full-scale range because it usually affects the output by the same percentage. To measure gain error, first determine the offset error, then set the digital inputs to all logic 1s, and finally measure the analog output. Ideally this output should be 1 LSB away from reference value. In terms of voltage,

$$\text{Gain error (\%)} = \left(\frac{V_{11} - V_{os}}{V_{ref} \left(1 - \frac{1}{2^n} \right)} - 1 \right) (100\%) \quad (14-4)$$

where V_{11} is the output voltage when the input digital code is all logic 1s, V_{os} is the offset error, and V_{ref} is the full-scale range (reference) value. Note the denominator is the ideal full-scale output voltage [see Eq. (14-3)].

Example 14-5

An 8-bit DAC wired for unipolar operation has a maximum gain error of $\pm 0.2\%$ of FSR. If the input reference is set at 5.12 V, what is the minimum output voltage when the input is all logic 1s? Assume the offset error is zero.

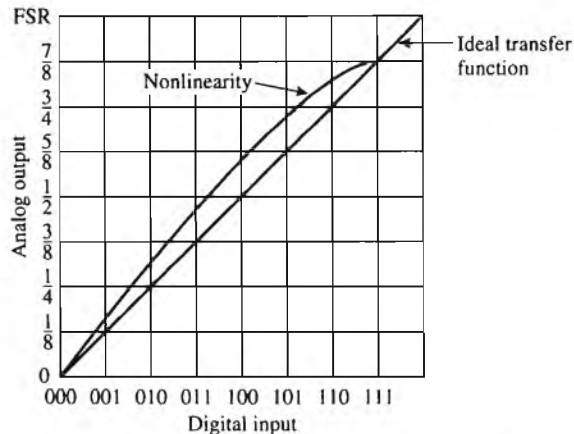


FIGURE 14-4 Nonlinearity error compared to the ideal transfer function.

Solution The denominator of Eq. (14-4) is 5.1 V [refer to Example 14-1 part (c)]. Substitute the values into Eq. (14-4).

$$-0.2\% = \left(\frac{V_{11} - 0}{5.1 \text{ V}} - 1 \right) \times 100\%$$

Solving for V_{11} yields $V_{11} = 5.0898 \text{ V}$. Note the output voltage cannot exceed 5.1 V.

14-1.4 Monotonic

A DAC is said to be monotonic if the analog output increases or remains the same as the digital input increases. This results in the output always being single-valued (an input digital code always produces one output value).

14-1.5 Relative Accuracy

Consider Fig. 14-1(a). The ideal transfer function is a straight line drawn from zero through full scale. Relative accuracy or integral nonlinearity (INL) is the maximum deviation at any input digital code from the ideal. See Fig. 14-4.

14-2 DIGITAL-TO-ANALOG CONVERSION PROCESS

14-2.1 Block Diagram

The block diagram for a resistive type of DAC is drawn in Fig. 14-5. Reference voltage V_{ref} is connected to a resistance network. A digital input code, via control circuitry, flips

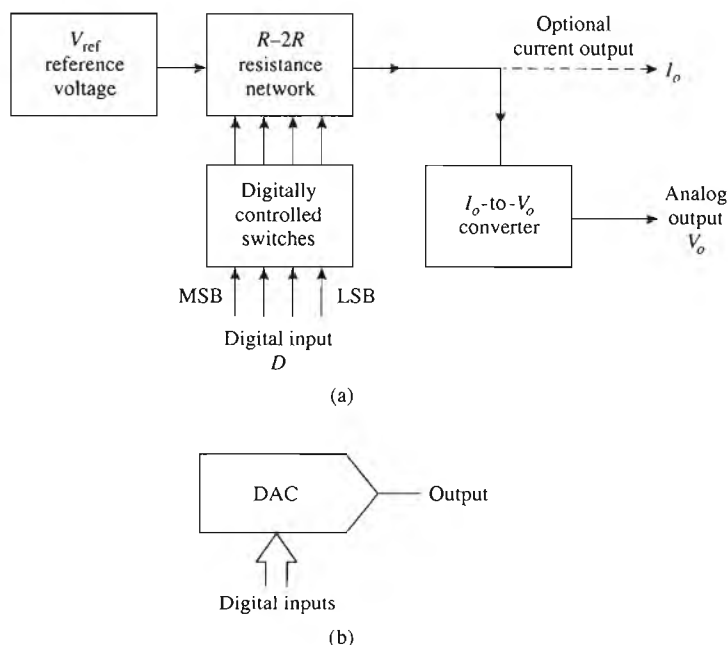


FIGURE 14-5 Block diagram and circuit symbol for a basic DAC.

switches (one for each bit) connected to the resistance network. The output of the resistance network is in the form of a current. This current may then be converted to a voltage. As previously mentioned, both current and voltage outputs are analog representations of the digital input code.

The actual digital-to-analog conversion takes place within the resistance network. Accordingly, we begin our study of DAC circuitry by looking next at the standard resistance network, called an R - $2R$ ladder network.

14-2.2 R - $2R$ Ladder Network

A 4-bit R - $2R$ ladder network is drawn in Fig. 14-6. Each digital input controls the position of its corresponding current switch. A current switch steers its ladder current either into a real ground (position 0) or a virtual ground (position 1). Thus the wiper of each switch is always at ground potential, so that the rung currents are constant except for the brief transition time of each switch.

In Fig. 14-6, rail currents flow horizontally; rung currents flow down through the bit switches. The rail current I_1 enters node 0, where it sees a resistance R_0 . R_0 is the equivalent resistance of a $2R$ resistor via switch D_0 to ground in parallel with a $2R$ terminate resistor. Thus $R_0 = 2R \parallel 2R = R$. As rail current I_1 leaves node 1, it sees R in series with $R_0 = R$ or $2R$. If we work our way back from the termination end to the voltage source, the value of resistance “looking” into a node is R . As shown in Fig. 14-6,

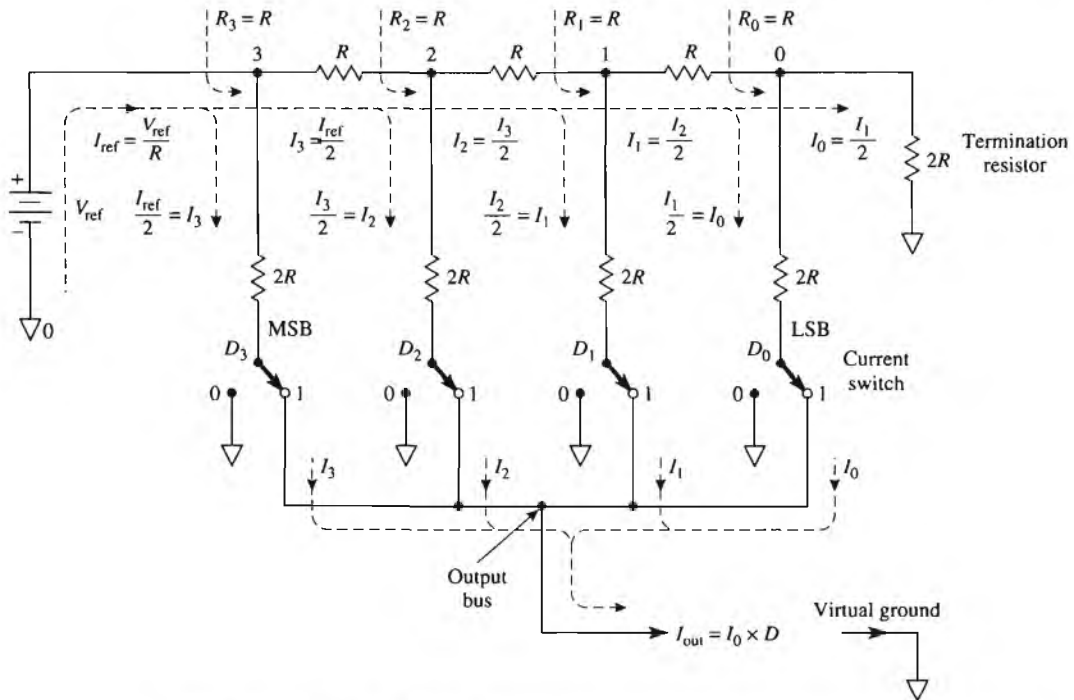


FIGURE 14-6 This R - $2R$ ladder converts the digital input code into an analog output current I_{out} .

$R_3 = R_2 = R_1 = R_0 = R$. R is called the *characteristic resistance* of the ladder network. In other words, V_{ref} sees the entire ladder network as a single resistor equal to R .

14-2.3 Ladder Currents

Since V_{ref} sees the ladder network as a resistance R , rail current I_{ref} is

$$I_{ref} = \frac{V_{ref}}{R} \quad (14-5)$$

The current pattern of the R - $2R$ network in Fig. 14-6 is analyzed as follows: The current I_{ref} splits into two equal parts at node 3: the rung current $I_3 = I_{ref}/2$ and the rail current $I_3 = I_{ref}/2$. Each rail current divides equally again at each node as it proceeds down the ladder. Rung currents are evaluated from

$$\begin{aligned} I_3 &= \frac{I_{ref}}{2} & I_2 &= \frac{I_3}{2} = \frac{I_{ref}}{4} \\ I_1 &= \frac{I_2}{2} = \frac{I_{ref}}{8} & I_0 &= \frac{I_1}{2} = \frac{I_{ref}}{16} \end{aligned} \quad (14-6)$$

I_0 is the current controlled by the LSB switch.

14-2.4 Ladder Equation

The output current bus receives current from a rung if the bit switch is in position 1. To write the output–input equation for the ladder network, we observe that I_{out} is the sum of all rung currents steered into the output bus by the bit switches. In equation form,

$$I_{\text{out}} = I_0 \times D \quad (14-7)$$

where D equals the decimal value of the digital input and I_0 is the smallest value of current in the ladder network. If we define I_0 as the resolution of the ladder, the output–input equation (14-7) can be expressed as

$$I_{\text{out}} = \text{resolution of 1 LSB} \times D \quad (14-8)$$

where

$$\text{current resolution of 1 LSB} = I_0 = \frac{I_{\text{ref}}}{2^n} = \frac{1}{2^n} \times \frac{V_{\text{ref}}}{R} \quad (14-9)$$

Example 14-6

The 4-bit resistance ladder of Fig. 14-6 has resistor values of $R = 10 \text{ k}\Omega$ and $2R = 20 \text{ k}\Omega$. V_{ref} equals 10 V. Find (a) the resolution of the ladder; (b) its output–input equation; (c) I_{out} for a digital input of 1111.

Solution (a) From Eq. (14-9),

$$\begin{aligned} \text{resolution of 1 LSB} &= \frac{1}{2^n} \times \frac{V_{\text{ref}}}{R} \\ &= \frac{1}{2^4} \times \frac{10 \text{ V}}{10 \text{ k}\Omega} \\ &= \frac{1}{16} \times 1 \text{ mA} = 62.5 \text{ }\mu\text{A} \end{aligned}$$

(b) Applying Eq. (14-8) yields

$$I_{\text{out}} = 62.5 \text{ }\mu\text{A} \times D$$

(c) The decimal value of binary 1111 is 15; therefore, $D = 15$ and

$$I_{\text{out}} = 62.5 \text{ }\mu\text{A} \times 15 = 0.9375 \text{ }\mu\text{A}$$

14-3 VOLTAGE OUTPUT DACs

As shown in Fig. 14-7, the output ladder current can be converted into a voltage by adding an op amp and a feedback resistor. Output voltage V_o is given by

$$V_o = -I_{\text{out}} R_f \quad (14-10a)$$

Substituting for I_{out} from Eq. (14-8) gives

$$V_o = -(\text{current resolution} \times D) \times R_f \quad (14-10b)$$

Rewrite Eq. (14-10b) as

$$V_o = -(\text{current resolution} \times R_f) \times D \quad (14-10c)$$

The coefficient of D is the voltage resolution and is given as

$$\text{voltage resolution} = I_o R_f \quad (14-10d)$$

and V_o can be written very simply as

$$V_o = -\text{voltage resolution} \times D \quad (14-10e)$$

In terms of the actual hardware, V_o is expressed as

$$V_o = -\left(\frac{V_{\text{ref}}}{R} \times \frac{1}{2^n} R_f\right) \times D \quad (14-11)$$

Example 14-7

For the voltage output DAC of Fig. 14-7, find (a) its voltage resolution of 1 LSB and (b) V_o when the digital input is 1111.

Solution (a) From Example 14-6, the value of $I_o = 62.5 \mu\text{A}$. From Eq. (14-10d),

$$\text{voltage resolution} = I_o R_f = 62.5 \mu\text{A} \times 10 \text{ k}\Omega = 0.625 \text{ V}$$

A 1-bit input change causes a 0.625-V output voltage change.

(b) Calculate the DAC's performance equation from Eq. (14-11).

$$V_o = -\left(\frac{10 \text{ V}}{10 \text{ k}\Omega} \times \frac{1}{2^4} \times 10 \text{ k}\Omega\right) \times D = -0.625 \text{ V} \times D$$

For a digital input of 1111, $D = 15$. Therefore,

$$V_o = -0.625 \text{ V} \times 15 = -9.375 \text{ V}$$

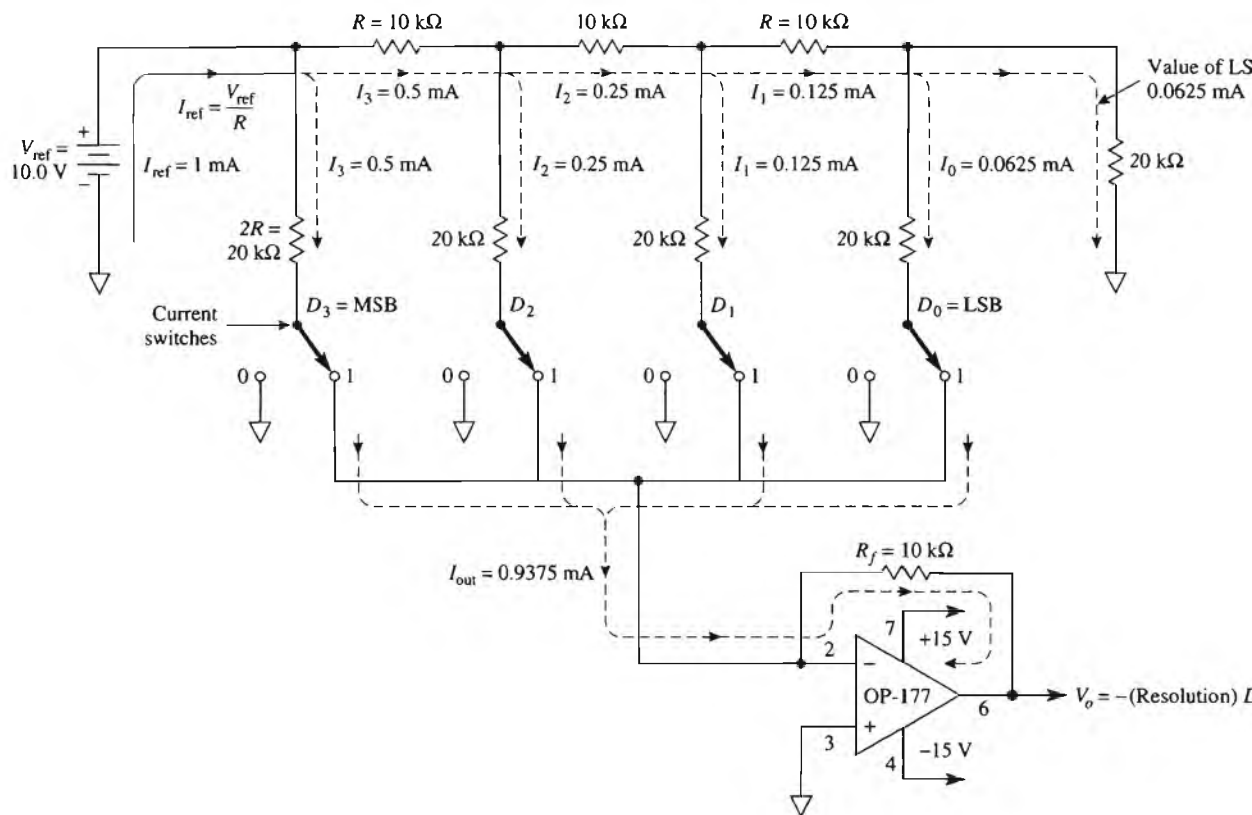


FIGURE 14-7 I_{out} of the R-2R ladder is converted into a voltage by the op amp and feedback resistor R_f .

14-4 MULTIPLYING DAC

Equation (14-11) can be rewritten to show how an MDAC or multiplying DAC operates.

$$V_o = (\text{constant}) \times V_{\text{ref}} \times D \quad (14-12)$$

where

$$\text{constant} = -\frac{R_f}{2^n R}$$

Equation (14-12) shows that V_o is the *product* of two input signals, V_{ref} and D , and *both* signals can be variables. One example for MDAC use is the volume control of an audio signal controlled by a microprocessor.

Suppose that V_{ref} is an audio signal that varies from 0 to 10 V in Fig. 14-7. From Example 14-7, if $D = 0001$, V_o would vary from 0 to 0.625 V. If the digital input is 1000, $D = 8$ and the volume of V_o would increase from 0 to $8 \times 0.625 \text{ V} = 5 \text{ V}$. Maximum volume of $15 \times 0.625 \text{ V} = 9.375 \text{ V}$ will occur when the digital input word is 1111, $D = 15$. Thus the MDAC performs as a digitally operated volume control.

14-5 8-BIT DIGITAL-TO-ANALOG CONVERTER: THE DAC-08

The DAC-08 is a low-cost, fast MDAC, housed in a 16-pin DIP. Its operating principles are examined by reference to the task performed by each of its terminals in Fig. 14-8.

14-5.1 Power Supply Terminals

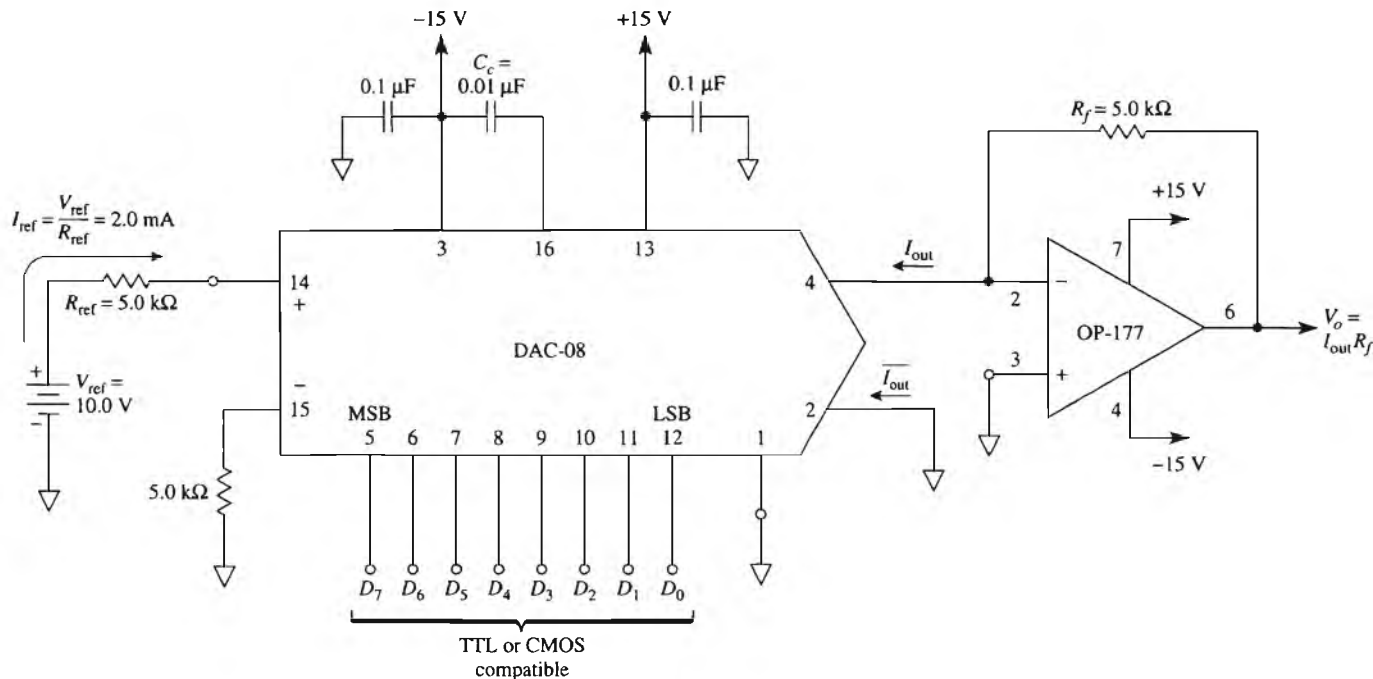
Pins 13 and 3 are the positive and negative supply terminals, respectively, and can have any value from ± 4.5 to $\pm 18 \text{ V}$. They should be bypassed with $0.1\text{-}\mu\text{F}$ and $0.01\text{-}\mu\text{F}$ capacitors, as shown in Fig. 14-8a.

14-5.2 Reference (Multiplying) Terminal

Flexibility of the DAC-08 is enhanced by having *two* rather than one reference input. Pins 14 and 15 allow positive or negative reference voltages, respectively. A positive reference voltage input is shown in Fig. 14-8a.

The user can adjust the DAC-08's input ladder current I_{ref} quite easily from $4 \mu\text{A}$ to 4 mA with a typical value of 2 mA .

$$I_{\text{ref}} = \frac{V_{\text{ref}}}{R_{\text{ref}}} \quad (14-13)$$



(a) DAC-08 wired for positive output voltages.

	Digital inputs								Analog output	
	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	I_{out}	V_o
LSB	0	0	0	0	0	0	0	1	7.812 μA	39 mV
Half scale	1	0	0	0	0	0	0	0	1.000 mA	5.0 V
Full scale	1	1	1	1	1	1	1	1	1.992 mA	9.96 V

(b) Summary of Examples 14-7 and 14-8.

FIGURE 14-8 An 8-bit DAC-08 is wired for unipolar voltage in (a). I_{out} has values given in (b) for three digital input words. The op amp converts I_{out} to voltage V_o .

14-5.3 Digital Input Terminals

Pins 5 through 12 identify the digital input terminals. Pin 5 is the most significant bit (MSB), D_7 . Pin 12 is the LSB, D_0 . The terminals are TTL or CMOS compatible. Logic input “0” is 0.8 V or less. Logic “1” is 2.0 V or greater, regardless of the power supply voltages. Usually, pin 1, V_{LC} , is grounded. However, it can be used to adjust the *logic input threshold* voltage V_{TH} according to $V_{TH} = V_{LC} + 1.4$ V. These digital inputs control eight internal current switches.

14-5.4 Analog Output Currents

Two current output terminals are provided in Fig. 14-8 to increase the DAC-08’s versatility. Pin 4 conducts output current I_{out} and pin 2 conducts its complement, $\overline{I_{out}}$. If an internal switch is positioned to “1,” its ladder rung current flows in the I_{out} bus. If positioned to “0,” ladder rung current flows in the $\overline{I_{out}}$ bus.

The current value of 1 LSB (resolution) is found from

$$\text{current resolution} = (\text{value of 1 LSB}) = \frac{V_{ref}}{R_{ref}} \times \frac{1}{2^n} \quad (14-14a)$$

I_{out} is calculated from

$$I_{out} = (\text{value of 1 LSB}) \times D \quad (14-14b)$$

where D is the decimal value of the digital input word. The full-scale output current in the pin 4 output bus occurs when the digital input is 11111111, so that $D = 255$. Let’s define this current as I_{fs} , where

$$I_{fs} = (\text{value of 1 LSB}) \times 255 \quad (14-15a)$$

The sum of all ladder rung currents in the DAC-08 equals I_{fs} . Since this sum always divides between the I_{out} and $\overline{I_{out}}$, the value of $\overline{I_{out}}$ is given by

$$\overline{I_{out}} = I_{fs} - I_{out} \quad (14-15b)$$

Example 14-8

Calculate (a) the ladder input current I_{ref} of the DAC-08 in Fig. 14-8; (b) the current value of 1 LSB.

Solution (a) From Eq. (14-13),

$$I_{ref} = \frac{10 \text{ V}}{5 \text{ k}\Omega} = 2 \text{ mA}$$

(b) From Eq. (14-14a),

$$\text{current value of 1 LSB} = \frac{10 \text{ V}}{5 \text{ k}\Omega} \times \frac{1}{2^8} = 7.812 \mu\text{A}$$

Example 14-9

For the DAC-08 circuit in Fig. 14-8, find the values of I_{out} and $\overline{I_{\text{out}}}$ when the digital input words are (a) 00000001; (b) 10000000; (c) 11111111.

Solution Example 14-8 showed that current output resolution is $7.812 \mu\text{A/bit}$. From Eq. (14-15a), evaluate I_{fs} .

$$I_{fs} = (\text{resolution})255 = 7.812 \mu\text{A} \times 255 = 1.992 \text{ mA}$$

The value of D is 1 for (a), 128 for (b), and 255 for (c). I_{out} can now be found from Eq. (14-14b):

$$(a) I_{\text{out}} = 7.812 \mu\text{A} \times 1 = 7.812 \mu\text{A} \text{ for } 00000001 \text{ input}$$

$$(b) I_{\text{out}} = 7.812 \mu\text{A} \times 128 = 1.000 \text{ mA for } 10000000 \text{ input}$$

$$(c) I_{\text{out}} = 7.812 \mu\text{A} \times 255 = 1.992 \text{ mA for } 11111111 \text{ input}$$

From Eq. (14-15b),

$$(a) \overline{I_{\text{out}}} = 1.992 \text{ mA} - 7.812 \mu\text{A} = 1.984 \text{ mA}$$

$$(b) \overline{I_{\text{out}}} = 1.992 \text{ mA} - 1.0 \text{ mA} = 0.992 \text{ mA}$$

$$(c) \overline{I_{\text{out}}} = 1.992 \text{ mA} - 1.992 \text{ mA} = 0$$

The results of Examples 14-8 and 14-9 are tabulated in Fig. 14-8(b).

14-5.5 Unipolar Output Voltage

In Fig. 14-8(a), the DAC-08's current output I_{out} is converted to an output voltage V_o by an external op amp and resistor R_f . The voltage output has a resolution of

$$\text{voltage resolution of 1 LSB} = \frac{V_{\text{ref}}}{R_{\text{ref}}} \times R_f \times \frac{1}{2^n} \quad (14-16a)$$

and V_o is given by

$$V_o = \text{voltage resolution} \times D = I_{\text{out}} R_f \quad (14-16b)$$

Example 14-10

For the DAC-08 circuit of Fig. 14-8(a), find V_o for digital inputs of (a) 00000001; (b) 11111111.

Solution From Eq. (14-16a),

$$\text{voltage resolution} = (10 \text{ V}) \frac{5 \text{ k}\Omega}{5 \text{ k}\Omega} \times \frac{1}{256} = 39.0 \text{ mV/bit}$$

(a) From Eq. (14-16b) with the value of $D = 1$,

$$V_o = 39.0 \text{ mV} \times 1 = 39.0 \text{ mV for } 00000001 \text{ input}$$

(b) The value of $D = 255$. From Eq. (14-16b),

$$V_o = 39.0 \text{ mV} \times 255 = 9.961 \text{ V for } 11111111 \text{ input}$$

14-5.6 Bipolar Analog Output Voltage

The versatility of the DAC-08 is shown by wiring it to give a bipolar analog output voltage in response to a digital input word [Fig. 14-9(a)]. The op amp and two resistors convert the *difference* between I_{out} and $\overline{I_{\text{out}}}$ into a voltage V_o :

$$V_o = (I_{\text{out}} - \overline{I_{\text{out}}})R_f \quad (14-17)$$

I_{out} drives V_o positive and $\overline{I_{\text{out}}}$ drives V_o negative. If the digital input word increases by 1 bit, I_{out} increases by 1 LSB. However, $\overline{I_{\text{out}}}$ must therefore *decrease* by 1 LSB. Therefore, the differential output current changes by 2 LSBs; thus we would expect the bipolar output voltage span to be twice that of a unipolar output (see Example 14-3).

V_{ref} has been increased slightly in Fig. 14-9(a) so that I_{ref} increases to 2.048 mA [Eq. (14-13)]. This increases the current value of 1 LSB to an even 8 μA [Eq. (14-14a)]. We show how the output voltage responds to digital inputs by an example.

Example 14-11

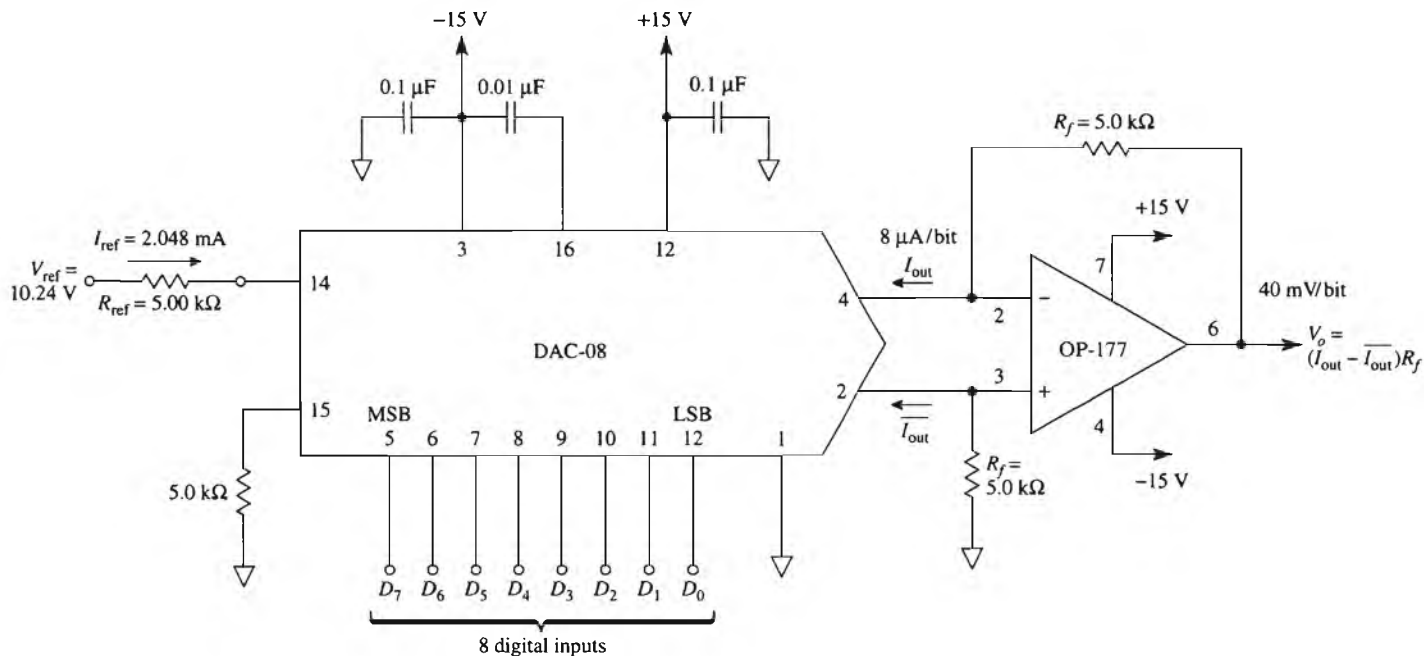
For the circuit of Fig. 14-9(a), calculate V_o for digital inputs of (a) 00000000; (b) 01111111; (c) 10000000; (d) 11111111.

Solution The current value of 1 LSB equals 8 μA . From Eq. (14-15a), $I_{fs} = (8 \mu\text{A})255 = 2.040 \text{ mA}$.

(a) From Eq. (14-14b), $I_{\text{out}} = (8 \mu\text{A}) \times 0 = 0$. Then from Eq. (14-15b), $\overline{I_{\text{out}}} = 2.040 \text{ mA} - 0 = 2.04 \text{ mA}$. Find V_o from Eq. (14-17):

$$V_o = (0 - 2.04 \text{ mA})(5 \text{ k}\Omega) = -10.20 \text{ V}$$

The values of I_{out} , $\overline{I_{\text{out}}}$, and V_o are calculated for (b), (c), and (d) and are summarized in Fig. 14-9(b).



(a) DAC-08 wired for bipolar output voltage.

	Digital inputs								Analog outputs		
	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	I_{out} (mA)	$\overline{I_{out}}$ (mA)	V_o (V)
Negative full scale	0	0	0	0	0	0	0	0	0	2.040	-10.20
Negative zero	0	1	1	1	1	1	1	1	1.016	1.024	-0.040
Positive zero	1	0	0	0	0	0	0	0	1.024	1.016	0.040
Positive full scale	1	1	1	1	1	1	1	1	2.040	0	10.20

(b) Tabulated solutions for Example 14-9.

FIGURE 14-9 The op amp converts complementary output currents of the DAC-08 into bipolar output voltages. The op amp is wired as a differential current-to-voltage converter.

Note that full-scale negative output voltage of -10.20 V occurs for an all-zero digital input. All ones give a positive full-scale output of plus 10.20 V . Note also that V_o never goes to precisely zero volts. When I_{out} is less than I_{out} by $8\text{ }\mu\text{A}$ (01111111), V_o equals -40 mV . Since this is the closest that V_o approaches to 0 V from negative full-scale, $V_o = -40\text{ mV}$ is called negative zero.

Many DACs must operate under the control of a microprocessor or computer. We therefore next present a microprocessor-compatible DAC.

14-6 MICROPROCESSOR COMPATIBILITY

14-6.1 Interfacing Principles

The programmer views the location of a DAC's register(s) or any other peripheral chip's registers as an address in the total memory space. DACs have "write only" registers. This means that a DAC has a register that the microprocessor can send binary digits to via the data bus. An ADC's registers are "read only" registers. These devices have registers whose contents can be "read" by the microprocessor via the data bus. Both DACs and ADCs have logic that permits selection via the address bus.

14-6.2 Memory Buffer Registers

"Read only" or "write only" registers have two operating states: *transparent* and *latching*. An idle register always remembers (latches) the last digital word written into it, and the register can be disconnected from the data bus. More specifically, the interface between data bus and register can be in the high-Z state, or essentially an open circuit.

When a register is *transparent*, it is connected to the data bus. For example, an 8-bit register in a DAC would allow its 8 data bits, D_7 through D_0 , to "read" the logic 1s and/or 0s present on each corresponding wire of the data bus placed there by the microprocessor (see Fig. 14-10).

How does the microprocessor locate one DAC, out of all other peripherals or memory addresses, that it is selected? This question is answered next.

14-6.3 The Selection Process

Every DAC register has an address just like any memory location of the microprocessor. To write to one particular DAC, the microprocessor places the address of that DAC on the address bus (see Fig. 14-10). One output of a local decoder goes low to enable a chip select, $\overline{\text{CS}}$, terminal on the selected DAC. The DAC's digital input buffer registers do not yet become transparent. The DAC is only partially selected.

To fully select the DAC, the microprocessor places a low on a line to the *chip enable*, CE, terminal. This line is controlled by the microprocessor's read/write line, which may also be referred to as MRW, MEMW, or R/W. When both the $\overline{\text{CS}}$ and CE terminals are low, only one DAC can communicate with the microprocessor. Its internal register

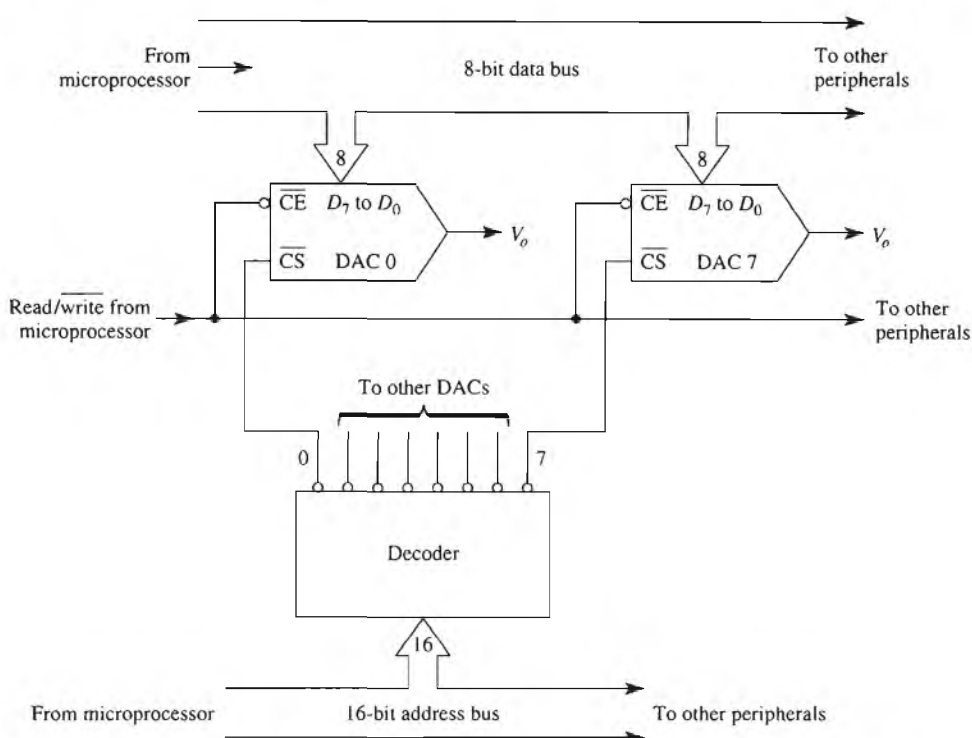


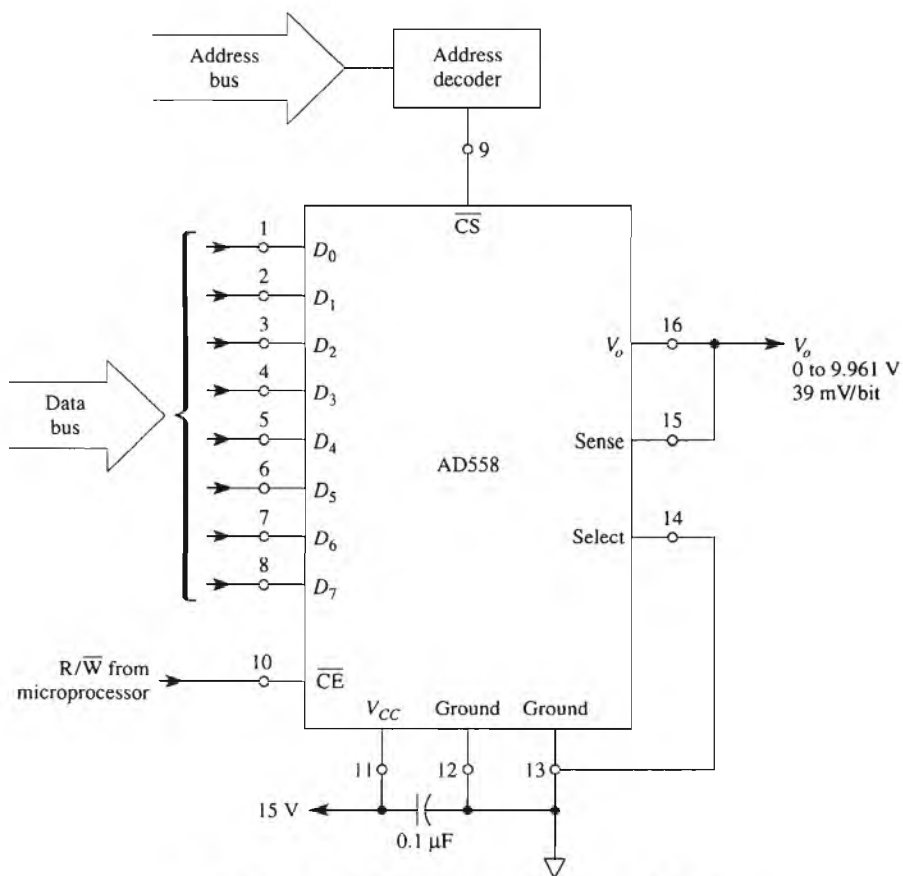
FIGURE 14-10 To select one DAC, the microprocessor places its address on the address bus. One output of the decoder goes low in response to its corresponding address code and enables the chip select terminal, $\overline{CS}$, of the DAC selected.

becomes transparent and accepts data from the data bus. The DAC immediately converts this digital data to an analog output voltage, V_o . This completes a “write only” operation. When either $\overline{CS}$ or $\overline{CE}$ goes high, the DAC’s register enters the latching state and remembers the last data written into it. Thus V_o is held at the analog equivalent voltage.

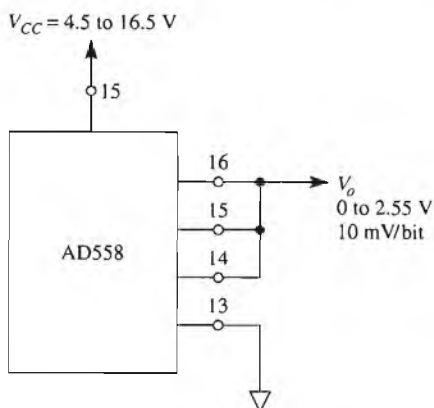
14-7 AD558 MICROPROCESSOR-COMPATIBLE DAC

14-7.1 Introduction

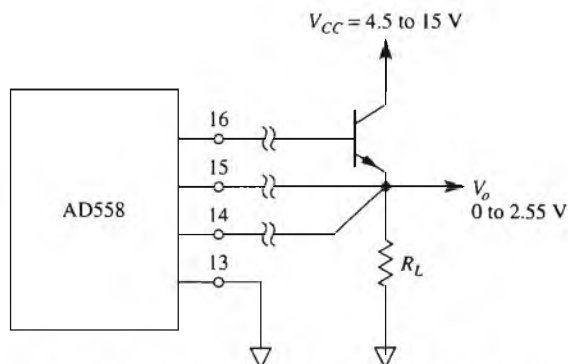
An example of a complete 8-bit microprocessor-compatible D/A converter is introduced in Fig. 14-11. The AD558 can operate continuously or it can be controlled by a microprocessor. It is complete with an on-board precision reference voltage, latching digital inputs, and select terminals. It also contains an op amp to give an analog output voltage that



(a) AD558 terminals pin programmed for $V_o = 0$ to 9.961 V.



(b) $V_o = 0$ to 2.55 V.



(c) Terminals 14 and 15 are extended to load R_L to eliminate the V_{BE} drop in the current boost transistor.

FIGURE 14-11 The AD588 is an 8-bit microprocessor-compatible D/A converter whose pinouts are shown in (a). It may be pin-programmed for 0 to 2.55 V as shown in (b). Terminal 15 and gain select terminal 14 may be extended to a load as shown in (c).

is pin-programmable for output ranges of 0 to 2.56 V or 0 to 10.0 V. Operation of the AD558 is studied by analyzing those tasks performed by its terminals.

14-7.2 Power Supply

Pin 11 is the power supply terminal V_{CC} in Fig. 14-11(a). It requires a minimum +4.5 V and has a maximum rating of +16.5 V. Pins 12 and 13 are the digital and analog grounds, respectively. This allows the user to maintain *separate* analog and digital grounds throughout a system, joining them at *only* one point. Usually, pins 12 and 13 are wired together and a 0.1- μ F bypass capacitor *must* be connected between V_{CC} and pin 12 or 13.

14-7.3 Digital Inputs

Pins 1 through 8 are the digital inputs D_0 to D_7 , with D_0 the LSB and D_7 the MSB. They are compatible with standard TTL or *low-voltage* CMOS. Logic 1 is 2.0 V minimum for a “1” bit. Logic 0 is 0.8 V maximum for a “0” bit.

The digital input pins connect the data bus to the AD558’s internal *memory latching register*, when the AD558 is selected. This condition is called *transparent*. When unselected, the latching register is essentially disconnected from the data bus and remembers the last word written into the latching register. This condition is called “latching.”

14-7.4 Logic Circuitry

The microprocessor executes a *write* command over the address bus via an address decoder and write line to the AD558’s logic control pins 9 and 10. They are called chip select ($\overline{CS}$) and chip enable ($\overline{CE}$), respectively.

If a “1” is present on either $\overline{CS}$ or $\overline{CE}$, the digital inputs are in the “latching” mode. They are disconnected from the data bus. The input latches remember the last word written by the microprocessor over the data bus. If both $\overline{CS}$ and $\overline{CE}$ are “0,” the AD558’s inputs are “transparent” and connect the input memory latch register to the data bus. The microprocessor can now write data into the DAC. Digital-to-analog conversion takes place immediately and is completed in about 200 ns.

14-7.5 Analog Output

As shown in Fig. 14-11, the analog output voltage (V_o) appears between pins 16 and 13 (analog ground). Pin 14 is called “select” (V_o gain). It is wired to pins 15 and 16 to set the output voltage range at 0 to 2.56 V, as in Fig. 14-11(b). The actual analog output range is 0 to 2.55 V, or 10 mV/bit for a digital input of 00000000 to 11111111. A 0-to-10-V output range connection is shown in Fig. 14-11a. The actual range is 0 to 9.961 V or 38.9 mV/bit (power supply voltage must *exceed* maximum V_o by 2 V

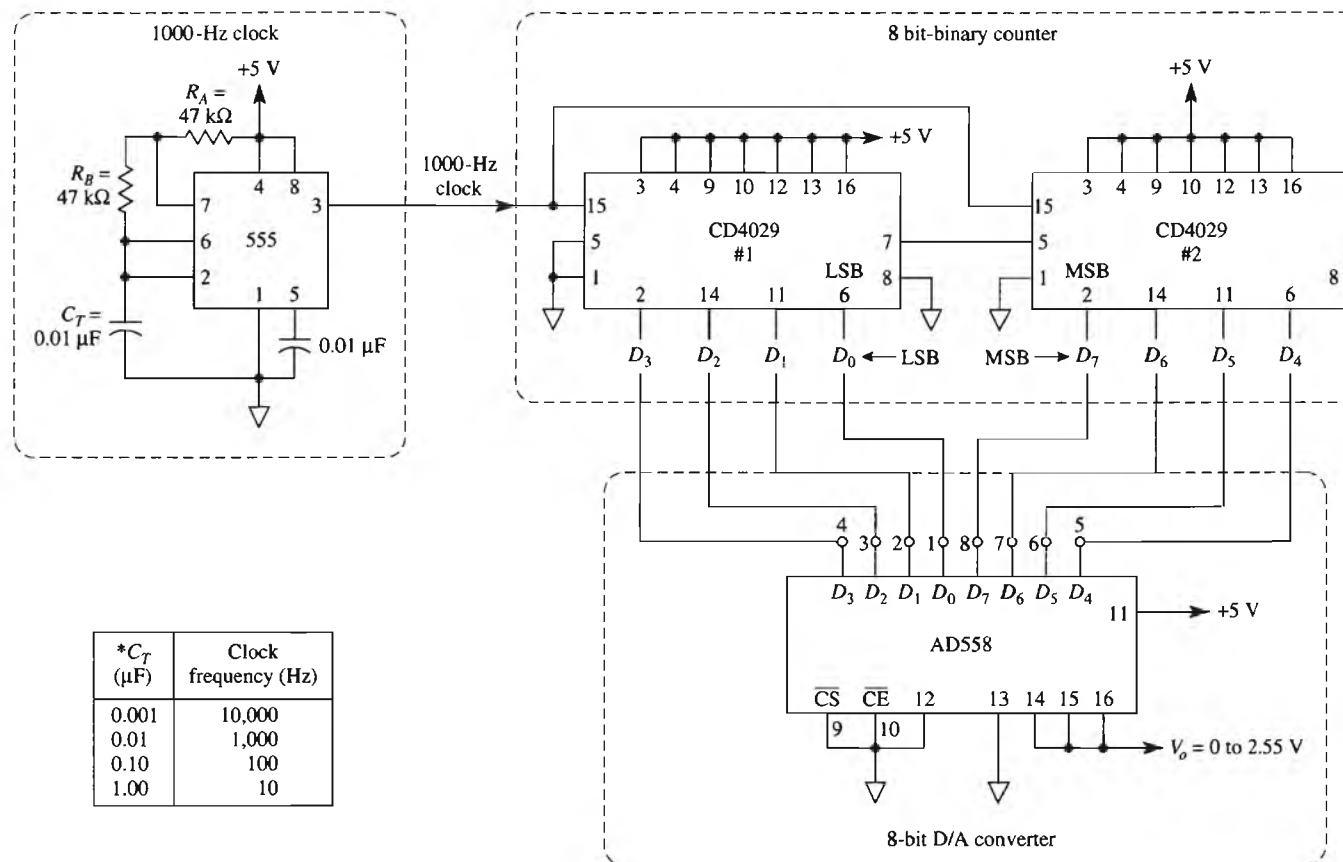


FIGURE 14-12 The 555 clock drives an 8-bit binary counter made from two CD4049 ICs. The outputs count in binary from 00000000 to 11111111 and then repeat. The digital count is converted by the DAC into an analog voltage that resembles a staircase.

minimum). Sense terminal 15 allows remote load-voltage sensing to eliminate the effects of IR drops in long leads to the load. It can also be used for current boost as in Fig. 14-11(c).

14-7.6 Dynamic Test Circuit

A single AD558 can be tested dynamically *without* a microprocessor by the stand-alone, low-parts-count test circuit of Fig. 14-12. Pins 9 and 10 of the AD558 are grounded. This connects the AD558's input register (transparent) to an 8-bit synchronous counter that simulates a data bus.

The test circuit consists of three ICs. One 555 timer is wired as a 1-kHz clock. It steps an 8-bit synchronous binary counter made from two CD4029s. The counter's outputs are wired to the digital inputs of the DAC. An oscilloscope is connected (dc coupled) to display V_o . It should be externally triggered from the negative edge of the MSB (pin 8 of the AD558 or pin 2 of the right CD4029 in Fig. 14-12). The analog voltage waveshape appearing at V_o will resemble a staircase.

Each clock pulse steps the counter up by one count and increases V_o by 10 mV. Thus the risers will equal 10 mV and the staircase will have 256 treads from 0 to 2.55 V. The tread on each step will occupy about 1 ms. Thus one staircase waveshape is generated every 256 ms. Any glitch, nonlinearity, or other abnormality will be quite apparent. The capacitor C_T can be changed to give faster or slower clock frequencies, as indicated in Fig. 14-12. Any visible glitches can be minimized but *not* eliminated from V_o . However, a *sample-and-hold* or *follow-and-hold amplifier* can be connected to V_o . It waits until the glitch settles down, samples V_o , and holds this correct value. The principle of sample-and-hold is presented in Chapter 15.

14-8 SERIAL DACs

14-8.0 Introduction

This section describes a 12-bit serial D/A converter. The data is shifted into the DAC one bit at a time usually from a microprocessor or a microcontroller. The authors selected the DAC-8043 from Analog Devices to show typical operation of serial DACs. This DAC is housed in an 8-pin mini-DIP. Figure 14-13 shows the pin designation as well as the functional block diagram. The device consists of a 12-bit serial-in to parallel-out shift register, a 12-bit DAC register, and a 12-bit digital-to-analog converter. It also contains the necessary control logic. Serial data is clocked into the input shift register, one bit on each rising edge of a clock signal. After 12 bits have been shifted in, the data is transferred to the DAC register by an active low state on the LD pin. Data in the DAC register is converted to an output current by the D/A converter. Since this particular device produces an output current, an external op amp must be connected to the DAC's output as shown in Fig. 14-14.

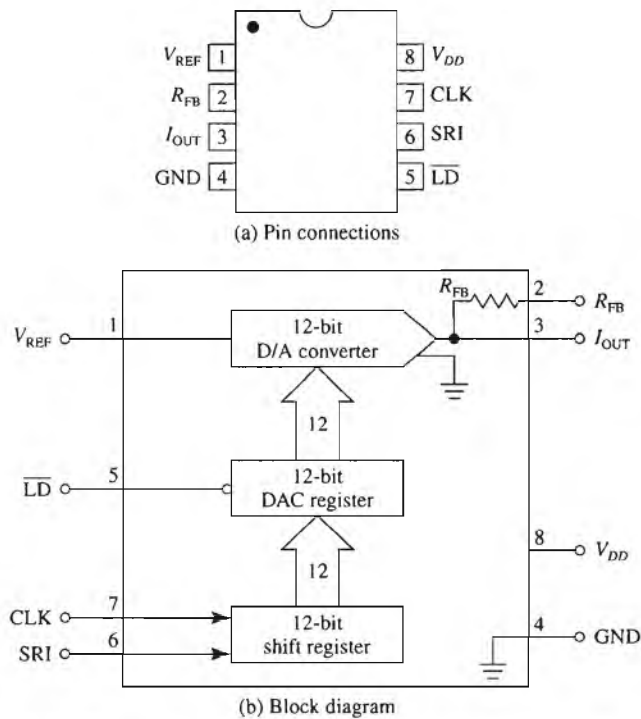


FIGURE 14-13 Pin designation and functional block diagram for a 12-bit serial DAC.

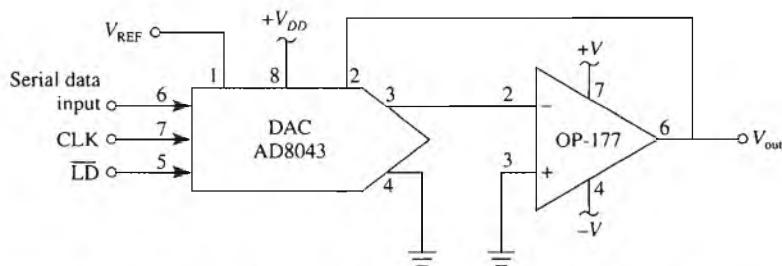


FIGURE 14-14 An op amp connected to the output of a serial DAC.

14-8.1 Interfacing a Serial DAC to a Microprocessor

Figure 14-15(a) shows a DAC-8043 interfaced to an MC6800 microprocessor. For this configuration the DAC-8043 is wired so that the input shift register is at address 0000_{hex}

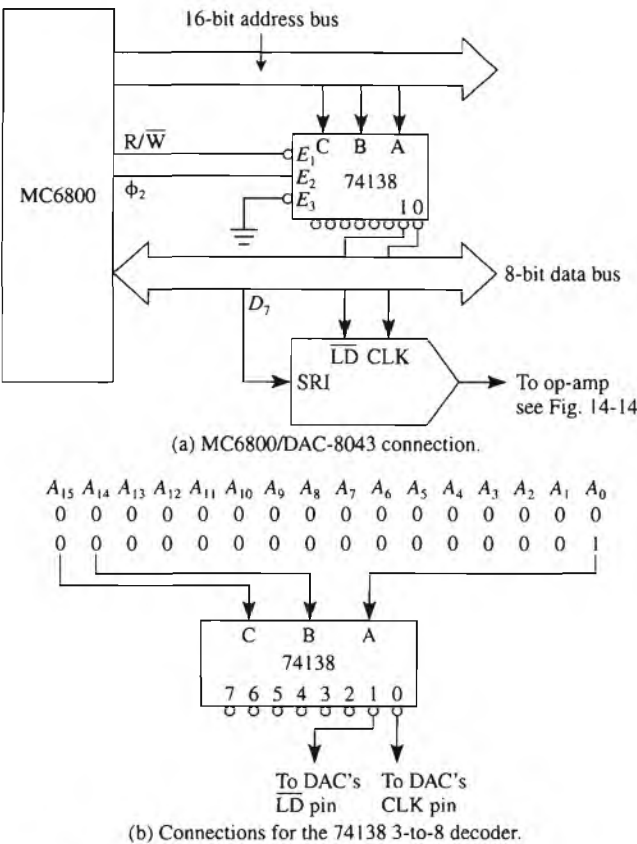


FIGURE 14-15 Interfacing the DAC-8043 serial D/A converter to the MC6800 microprocessor.

and the data is transferred to the DAC register when address 0001_{hex} is sent by the μP . A more detailed schematic of the decoder chip (74138) and the microprocessor's address lines are shown in Fig. 14-15(b).

14-8.2 Assembly Language Programming

Note that the 12 bits of data to be sent to the DAC are stored at memory locations $D000_{\text{hex}}$ and $D001_{\text{hex}}$ as shown in Fig. 14-16. For this example, the data to be sent are all logic 1s. The Xs in Fig. 14-16 indicate don't care bits and are not considered valid data for this application; they are not to be sent. The following subroutine will rotate the data at location

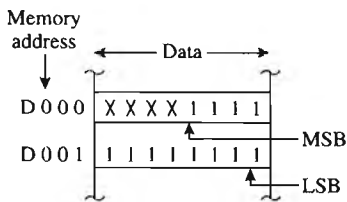


FIGURE 14-16 Data to be sent is stored in memory.

D000 so that the MSB will be sent first. The following assembly language code is written as a subroutine that is called to send the data one bit at a time to the DAC on data bus line D₇.

Subroutine Code

	LDAB	#04	}	Initialize accumulators A and B
	LDAA	\$D000		
LOOP	LSLA		}	Shift don't care data, Xs, out of the way
	DECB			
	BNE	LOOP		
	LDAB	#04	}	Reinitialize count value, branch to SHIFT subroutine
	BSR	SHIFT		
	LDAB	#08	}	Reinitialize to send next 8 bits
	LDAA	\$D001		
	BSR	SHIFT		
	STAA	\$0001	}	Transfer data from shift reg. to DAC reg. Return from subroutine.
	RTS			
SHIFT	STAA	\$0000	}	Subroutine to shift data on data bus line D ₇ .
	LSLA			
	DECB			
	BNE	SHIFT		
	RTS			

The 74138 IC in Fig. 14-15 could be used to interface three more serial DACs to the microprocessor.

PROBLEMS

14-1. What is the resolution for (a) 10-bit DAC and (b) 14-bit DAC?

14-2. If the input reference voltage is 10 V for each of the DACs of Problem 14-1, what is the equivalent least significant voltage value?

- 14-3. What is the maximum output voltage for a 10-bit DAC if the reference voltage is set at 10.24 V?
- 14-4. A 12-bit DAC has a least significant bit resolution of 2.5 mV/bit, what is the output voltage if the input binary patterns are:
(a) 100011001000
(b) 001100100101
The most significant bit is the leftmost bit.
- 14-5. The output voltage of an 8-bit DAC is 1.9 V. What is the input digital code if the reference voltage is 5.12 V?
- 14-6. A 12-bit DAC has an offset error of $-1/2$ LSB and a gain error of $-1/2$ LSB. If the input reference voltage is 10.24 V, what is the output voltage when the input digital code is all 1s?
- 14-7. Refer to the R-2R ladder network of Fig. 14-7. Let $V_{\text{ref}} = 10$ V, $R_f = 5$ k Ω , $R = 5$ k Ω , and $2R = 10$ k Ω . Find (a) characteristic ladder resistance, (b) I_0 ; (c) voltage resolution; (d) output-input equation; (e) V_o .
- 14-8. What is the voltage at pins 2 and 3 of the op amp in Fig. 14-9(a) when the digital inputs are 1000000? Also calculate the voltage across the feedback resistor to get V_o .
- 14-9. A basic DAC consists of a reference voltage, ladder network, current switches, and op amp. Name two additional features required to make the DAC microprocessor compatible.
- 14-10. Can the AD558 and DAC-08 be used as multiplying DACs?
- 14-11. These questions refer to the AD558. (a) Name the terminals that allow this DAC to be selected. (b) Describe the digital input register's latching mode and (c) transparent mode of operation.
- 14-12. (a) What is the output-input equation for an AD558 DAC?
(b) Find V_o for an input code of 10000000.
- 14-13. If the 74138 decoder chip shown in Fig. 14-15 is used to interface three more serial DACs, at what address would their registers be located?